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# WMA11/01 January 2022 Worked Solutions

Jan 2022 paper rebuilt with the worked solution after each question.

**10**
paper questions

**WMA11**
Pure 1

**Single-paper solutions**
standalone IAL route

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**Prepared for Dr Eslam Ahmed - eliteigcse.com**

P1

**PAST PAPER**

# WMA11/01 January 2022

**Jan 2022 | 10 questions | 75 marks**

**10**
questions

**75**
marks

**Answers**
worked solution after each question

# Question 1

4 marks

Integration

**1.** Find

$$\int\left(\frac{8x^3}{5}-\frac{2}{3x^4}-1\right)\mathrm{d}x$$

giving each term in simplest form.

**(4)**

**(Total 4 marks)**

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# Worked Solution - Question 1

Integration

Topic group

**1. Rewrite using powers**

$$\frac{8x^3}{5} - \frac{2}{3x^4} - 1 = \frac{8}{5}x^3 - \frac{2}{3}x^{-4} - 1.$$

**2. Integrate term by term**

$\int \frac{8}{5}x^3 dx = \frac{2}{5}x^4$, $\int -\frac{2}{3}x^{-4} dx = \frac{2}{9}x^{-3}$, and $\int -1 dx = -x.$

**3. Write in simplest form**

Therefore the integral is $\frac{2x^4}{5} + \frac{2}{9x^3} - x + c.$

**Final answer**

$$\frac{2x^4}{5} + \frac{2}{9x^3} - x + c.$$

# Question 2

6 marks

Quadratics

**2.**

$$f(x) = 11 - 4x - 2x^2$$

(a) Express f($x$) in the form

$$a + b(x + c)^2$$

where $a$, $b$ and $c$ are integers to be found.

**(3)**

(b) Sketch the graph of the curve $C$ with equation $y = f(x)$, showing clearly the coordinates of the point where the curve crosses the $y$-axis.

**(2)**

(c) Write down the equation of the line of symmetry of $C$.

**(1)**

**(Total 6 marks)**

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# Worked Solution - Question 2

Quadratics

Topic group

### 1. Complete the square

$f(x) = 11 - 4x - 2x^2 = -2(x^2 + 2x) + 11.$

### 2. Write in vertex form

$-2(x^2 + 2x) + 11 = -2\big((x+1)^2 - 1\big) + 11 = 13 - 2(x+1)^2.$

### 3. Identify the graph features

The graph is a downward-opening parabola with maximum point $(-1, 13)$. The y-intercept is $f(0) = 11$, so it crosses the y-axis at $(0, 11)$.

### 4. State the line of symmetry

From $13 - 2(x+1)^2$, the line of symmetry is $x = -1$.

**Final answer**

$(a)\ 13 - 2(x+1)^2.$

$(b)\ y\text{-intercept } (0, 11).$

$(c)\ x = -1.$

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# Question 3

7 marks

Quadratics

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**3.** **In this question you must show all stages of your working.**

**Solutions relying on calculator technology are not acceptable.**

(i)

$$\mathrm{f}(x) = \left(x + \sqrt{2}\right)^2 + \left(3x - 5\sqrt{8}\right)^2$$

Express $\mathrm{f}(x)$ in the form $ax^2 + bx\sqrt{2} + c$ where $a$, $b$ and $c$ are integers to be found.

**(3)**

(ii) Solve the equation

$$\sqrt{3}\left(4y - 3\sqrt{3}\right) = 5y + \sqrt{3}$$

giving your answer in the form $p + q\sqrt{3}$ where $p$ and $q$ are simplified fractions to be found.

**(4)**

**(Total 7 marks)**

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# Worked Solution - Question 3

Quadratics

Topic group

### 1. Expand the first square

$(x+\sqrt{2})^2 = x^2 + 2\sqrt{2}x + 2.$

### 2. Expand the second square

Since $\sqrt{8} = 2\sqrt{2}$, $(3x - 5\sqrt{8})^2 = (3x - 10\sqrt{2})^2 = 9x^2 - 60\sqrt{2}x + 200.$

### 3. Collect terms

$f(x) = x^2 + 2\sqrt{2}x + 2 + 9x^2 - 60\sqrt{2}x + 200 = 10x^2 - 58\sqrt{2}x + 202.$

### 4. Expand the surd equation

$\sqrt{3}(4y - 3\sqrt{3}) = 5y + \sqrt{3}$ gives $4\sqrt{3}y - 9 = 5y + \sqrt{3}.$

### 5. Solve for y

$y(4\sqrt{3} - 5) = 9 + \sqrt{3}$, so $y = \dfrac{9+\sqrt{3}}{4\sqrt{3}-5}.$

### 6. Rationalise the denominator

$$y = \frac{(9+\sqrt{3})(4\sqrt{3}+5)}{(4\sqrt{3})^2 - 5^2} = \frac{57 + 41\sqrt{3}}{23} = \frac{57}{23} + \frac{41}{23}\sqrt{3}.$$

**Final answer**

$(i)\ f(x) = 10x^2 - 58\sqrt{2}x + 202. \quad (ii)\ y = \dfrac{57}{23} + \dfrac{41}{23}\sqrt{3}.$

# Question 4

7 marks

Graphs of Functions

**4.**

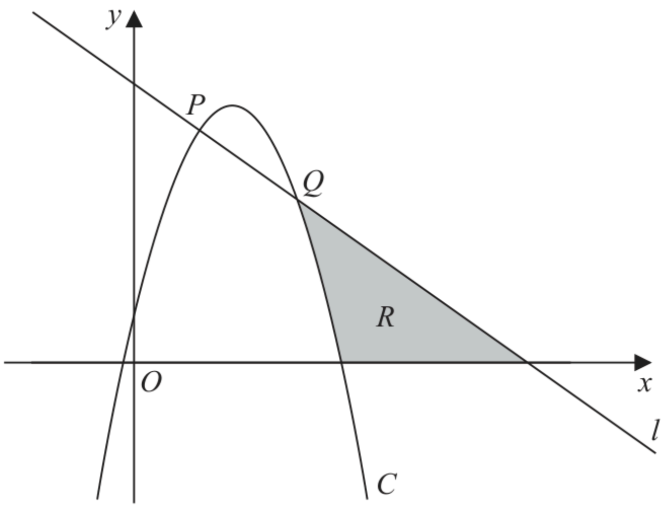


**Figure 1**

**In this question you must show all stages of your working.**

**Solutions relying on calculator technology are not acceptable.**

Figure 1 shows a line $l$ with equation $x + y = 6$ and a curve $C$ with equation $y = 6x - 2x^2 + 1$

The line $l$ intersects the curve $C$ at the points $P$ and $Q$ as shown in Figure 1.

(a) Find, using algebra, the coordinates of $P$ and the coordinates of $Q$.

**(4)**

The region $R$, shown shaded in Figure 1, is bounded by $C$, $l$ and the $x$-axis.

(b) Use inequalities to define the region $R$.

**(3)**

**(Total 7 marks)**

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# Worked Solution - Question 4

Graphs of Functions

Topic group

### 1. Set line equal to curve

The line $x + y = 6$ has equation $y = 6 - x$. Intersections satisfy $6 - x = 6x - 2x^2 + 1$.

### 2. Solve the quadratic

Rearranging gives $2x^2 - 7x + 5 = 0$, so $(2x - 5)(x - 1) = 0$. Therefore $x = 1$ or $x = \frac{5}{2}$.

### 3. Find the coordinates

Using $y = 6 - x$, the points are $P = (1, 5)$ and $Q = \left(\frac{5}{2}, \frac{7}{2}\right)$.

### 4. Read the bounded region

Region $R$ is above the x-axis, below the line, and above the curve.

### 5. Write the inequalities

Thus $y \geq 0$, $y \leq 6 - x$, and $y \geq 6x - 2x^2 + 1$.

**Final answer**

$(a)\ P = (1, 5),\ Q = \left(\frac{5}{2}, \frac{7}{2}\right)$.

$(b)\ y \geq 0,\ y \leq 6 - x,\ y \geq 6x - 2x^2 + 1$.

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# Question 5

7 marks

Radians

**5.**

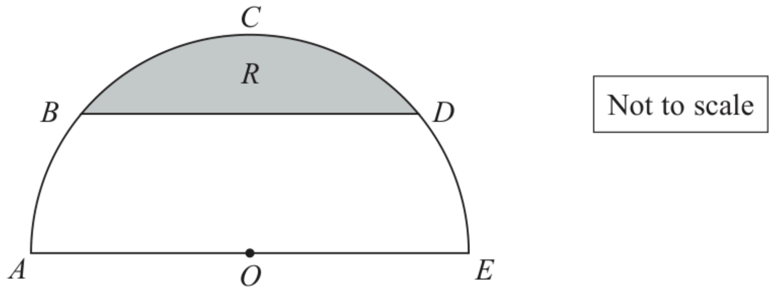


**Figure 2**

Figure 2 shows a plan view of a semicircular garden *ABCDEOA*

The semicircle has

- centre *O*
- diameter *AOE*
- radius 3 m

The straight line *BD* is parallel to *AE* and angle *BOA* is 0.7 radians.

(a) Show that, to 4 significant figures, angle *BOD* is 1.742 radians.

**(1)**

The flowerbed *R*, shown shaded in Figure 2, is bounded by *BD* and the arc *BCD*.

(b) Find the area of the flowerbed, giving your answer in square metres to one decimal place.

**(3)**

(c) Find the perimeter of the flowerbed, giving your answer in metres to one decimal place.

**(3)**

**(Total 7 marks)**

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# Worked Solution - Question 5

Radians

Topic group

### 1. Find angle BOD

Since $BD$ is parallel to the diameter $AE$, the two side angles at the centre are both $0.7$ radians. Therefore $\angle BOD = \pi - 0.7 - 0.7 = \pi - 1.4 = 1.74159\ldots$, so $1.742$ radians to 4 significant figures.

### 2. Find the sector area

With $r = 3$ and $\theta = 1.74159\ldots$, sector area $= \dfrac{1}{2}r^2\theta = \dfrac{1}{2}(3)^2(1.74159\ldots) = 7.837\ldots$.

### 3. Subtract triangle BOD

Triangle area $= \dfrac{1}{2}(3)(3)\sin(1.74159\ldots) = 4.434\ldots$. The segment area is $7.837\ldots - 4.434\ldots = 3.403\ldots$, so $3.4\ \mathrm{m}^2$.

### 4. Find the arc length

Arc $BCD = r\theta = 3(1.74159\ldots) = 5.225\ldots$.

### 5. Find the chord and perimeter

$BD = 2r\sin\dfrac{\theta}{2} = 6\sin(0.87079\ldots) = 4.590\ldots$. Perimeter $= 5.225\ldots + 4.590\ldots = 9.8\ \mathrm{m}$.

**Final answer**

$(a)\ \angle BOD = 1.742\ \mathrm{rad}.$

$(b)\ 3.4\ \mathrm{m}^2.$

$(c)\ 9.8\ \mathrm{m}.$

# Question 6

11 marks

Integration

**6.** The curve $C$ has equation $y = \mathrm{f}(x)$ where $x > 0$

Given that

- $\mathrm{f}'(x) = \dfrac{(x+3)^2}{x\sqrt{x}}$
- the point $P(4, 20)$ lies on $C$

(a) (i) find the value of the gradient at $P$

(ii) Hence find the equation of the tangent to $C$ at $P$, giving your answer in the form $ax + by + c = 0$ where $a$, $b$ and $c$ are integers to be found.

**(4)**

(b) Find $\mathrm{f}(x)$, simplifying your answer.

**(7)**

**(Total 11 marks)**

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# Worked Solution - Question 6

Integration

Topic group

**1. Simplify f'(x)**

$$f'(x)=\frac{(x+3)^2}{x\sqrt{x}}=\frac{x^2+6x+9}{x^{3/2}}=x^{1/2}+6x^{-1/2}+9x^{-3/2}.$$

**2. Find the gradient at P**

At $x=4$, $f'(4)=2+3+\frac{9}{8}=\frac{49}{8}$.

**3. Find the tangent equation**

Using $P(4,20)$, $y-20=\frac{49}{8}(x-4)$. Multiplying out gives $49x-8y-36=0$.

**4. Integrate f'(x)**

$$f(x)=\int\left(x^{1/2}+6x^{-1/2}+9x^{-3/2}\right)dx=\frac{2}{3}x^{3/2}+12x^{1/2}-18x^{-1/2}+c.$$

**5. Use P(4,20)**

$20=\frac{2}{3}(8)+12(2)-\frac{18}{2}+c=\frac{61}{3}+c$, so $c=-\frac{1}{3}$.

**6. Write f(x)**

$$f(x)=\frac{2}{3}x\sqrt{x}+12\sqrt{x}-\frac{18}{\sqrt{x}}-\frac{1}{3}.$$

**Final answer**

$(a)(i)\ \frac{49}{8}.\quad (a)(ii)\ 49x-8y-36=0.$

$(b)\ f(x)=\frac{2}{3}x\sqrt{x}+12\sqrt{x}-\frac{18}{\sqrt{x}}-\frac{1}{3}.$

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# Question 7

11 marks

Differentiation

**7.**

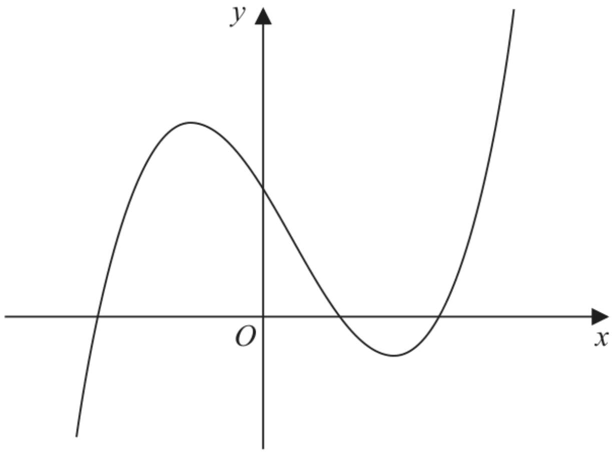


**Figure 3**

Figure 3 shows a sketch of part of the curve with equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (x + 4)(x - 2)(2x - 9)$$

Given that the curve with equation $y = \mathrm{f}(x) - p$ passes through the point with coordinates (0, 50)

(a) find the value of the constant $p$.

**(2)**

Given that the curve with equation $y = \mathrm{f}(x + q)$ passes through the origin,

(b) write down the possible values of the constant $q$.

**(2)**

(c) Find $\mathrm{f}'(x)$.

**(4)**

(d) Hence find the range of values of $x$ for which the gradient of the curve with equation $y = \mathrm{f}(x)$ is less than $-18$

**(3)**

**(Total 11 marks)**

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# Worked Solution - Question 7

Differentiation

Topic group

### 1. Find p from the translated curve

$f(0) = (4)(-2)(-9) = 72$. Since $y = f(x) - p$ passes through $(0, 50)$, $72 - p = 50$, so $p = 22$.

### 2. Use roots for f(x + q)

If $y = f(x + q)$ passes through the origin, then $f(q) = 0$. From $f(x) = (x + 4)(x - 2)(2x - 9)$, the possible values are $q = -4, 2, \frac{9}{2}$.

### 3. Expand f(x)

$(x + 4)(x - 2)(2x - 9) = 2x^3 - 5x^2 - 34x + 72$.

### 4. Differentiate

$f'(x) = 6x^2 - 10x - 34$.

### 5. Set the gradient less than -18

$6x^2 - 10x - 34 < -18$, so $6x^2 - 10x - 16 < 0$, or $3x^2 - 5x - 8 < 0$.

### 6. Solve the inequality

$3x^2 - 5x - 8 = (3x - 8)(x + 1)$. Since the quadratic opens upward, it is negative between the roots: $-1

**Final answer**

$(a)\ p = 22.$

$(b)\ q = -4, 2, \dfrac{9}{2}.$

$(c)\ f'(x) = 6x^2 - 10x - 34.$

$(d)\ -1

# Question 8

9 marks

Straight Line

**8.** The line $l_1$ has equation

$$2x - 5y + 7 = 0$$

(a) Find the gradient of $l_1$

**(1)**

Given that

- the point $A$ has coordinates $(6, -2)$
- the line $l_2$ passes through $A$ and is perpendicular to $l_1$

(b) find the equation of $l_2$ giving your answer in the form $y = mx + c$, where $m$ and $c$ are constants to be found.

**(3)**

The lines $l_1$ and $l_2$ intersect at the point $M$.

(c) Using algebra and showing all your working, find the coordinates of $M$.

(*Solutions relying on calculator technology are not acceptable.*)

**(3)**

Given that the diagonals of a square $ABCD$ meet at $M$,

(d) find the coordinates of the point $C$.

**(2)**

**(Total 9 marks)**

# Worked Solution - Question 8

Straight Line

Topic group

### 1. Find the gradient of l1

$2x - 5y + 7 = 0$ gives $y = \frac{2}{5}x + \frac{7}{5}$, so the gradient is $\frac{2}{5}$.

### 2. Find l2

The gradient perpendicular to $\frac{2}{5}$ is $-\frac{5}{2}$. Through $A(6, -2)$, $y + 2 = -\frac{5}{2}(x - 6)$, so $y = -\frac{5}{2}x + 13$.

### 3. Find M

At the intersection, $\frac{2}{5}x + \frac{7}{5} = -\frac{5}{2}x + 13$. Multiplying by $10$ gives $4x + 14 = -25x + 130$, so $x = 4$.

### 4. Find the y-coordinate of M

$y = \frac{2}{5}(4) + \frac{7}{5} = 3$, so $M = (4, 3)$.

### 5. Use midpoint of the diagonal

In a square, the diagonals bisect each other, so $M$ is the midpoint of $AC$. Thus $C = 2M - A = (8, 6) - (6, -2) = (2, 8)$.

**Final answer**

$(a)$ $\frac{2}{5}$.

$(b)$ $y = -\frac{5}{2}x + 13$.

$(c)$ $M = (4, 3)$.

$(d)$ $C = (2, 8)$.

## Question 9

**9.**

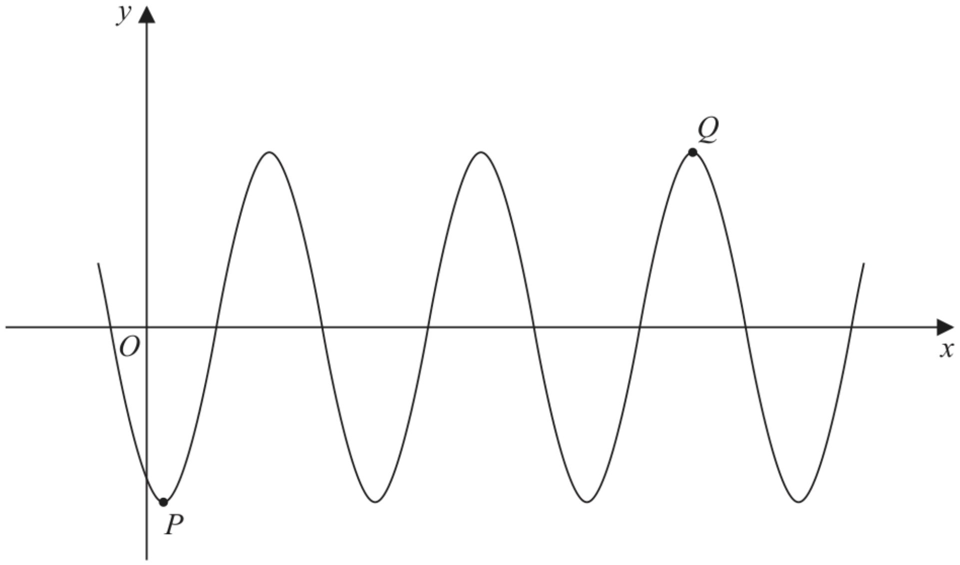


**Figure 4**

Figure 4 shows part of the curve with equation

$$y = A\cos(x - 30)°$$

where $A$ is a constant.

The point $P$ is a minimum point on the curve and has coordinates $(30, -3)$ as shown in Figure 4.

(a) Write down the value of $A$.

**(1)**

The point $Q$ is shown in Figure 4 and is a maximum point.

(b) Find the coordinates of $Q$.

**(3)**

**(Total 4 marks)**

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# Worked Solution - Question 9

Trigonometric Functions

Topic group

### 1. Use the minimum point

At $P(30, -3)$, $x - 30 = 0$, so $\cos 0° = 1$. Therefore $A = -3$.

### 2. Find maximum points

Since $A = -3$, maximum points occur when $\cos(x - 30)° = -1$.

### 3. Use the shown maximum Q

$x - 30 = 180 + 360n$, so $x = 210 + 360n$. The maximum shown at $Q$ is two full periods after the first maximum, so $x = 210 + 720 = 930$.

### 4. Write Q

The maximum y-value is $3$, hence $Q = (930, 3)$.

**Final answer**

$(a)$ $A = -3.$

$(b)$ $Q = (930, 3).$

# Question 10

9 marks

Graphs of Functions

**10.** The curve $C$ has equation

$$y = \frac{1}{x^2} - 9$$

(a) Sketch the graph of $C$.

On your sketch

- show the coordinates of any points of intersection with the coordinate axes
- state clearly the equations of any asymptotes

**(4)**

The curve $D$ has equation $y = kx^2$ where $k$ is a constant.

Given that $C$ meets $D$ at 4 distinct points,

(b) find the range of possible values for $k$.

**(5)**

**(Total 9 marks)**

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# Worked Solution - Question 10

Graphs of Functions

Topic group

### 1. Find intercepts

For x-intercepts, set $\dfrac{1}{x^2} - 9 = 0$. Then $x^2 = \dfrac{1}{9}$, so $x = \pm\dfrac{1}{3}$. There is no y-intercept because $x = 0$ is not in the domain.

### 2. State asymptotes

The vertical asymptote is $x = 0$. As $|x|$ becomes large, $\dfrac{1}{x^2} \to 0$, so the horizontal asymptote is $y = -9$.

### 3. Set up intersections with D

For intersections with $y = kx^2$, solve $\dfrac{1}{x^2} - 9 = kx^2$.

### 4. Use u = x squared

Let $u = x^2$, where $u > 0$. Multiplying by $x^2$ gives $1 - 9u = ku^2$, so $ku^2 + 9u - 1 = 0$.

### 5. Require two positive u-values

Four distinct x-values require two distinct positive values of $u$, since each positive $u$ gives $x = \pm\sqrt{u}$.

### 6. Find the range of k

For two positive roots, the product $-\dfrac{1}{k}$ and sum $-\dfrac{9}{k}$ must both be positive, so $k < 0$. Also the discriminant must be positive: $81 + 4k > 0$, so $k > -\dfrac{81}{4}$.

Therefore $-\dfrac{81}{4}

**Final answer**

$(a)$ $x$-intercepts $\left(-\dfrac{1}{3},0\right),\left(\dfrac{1}{3},0\right)$; asymptotes $x=0,\ y=-9.$

$(b)\ -\dfrac{81}{4}